# Today

- Finite square well

## 2.3 Finite square well

**Question 2.3.1:** Considering a finite square well of the same width as the infinite square well with $V(x) = 0$ for $-\frac{a}{2} \leq x \leq \frac{a}{2}$ and $V(x) = -V_0$ for $|x| > \frac{a}{2}$, which of the following are true about bound states in the finite square well? (A). The wavefunction is nonzero outside the well. (B). The ground state is symmetric with no nodes. (C). $\psi_n(x)$ has the same number of nodes as the nodes in the interior for $\psi_n(x)$ of the infinite square well. (D). $E_n$ is smaller than $E_n$ of the infinite square well. (E). There are a finite number of bound states.

**Answer:** All are correct. Because the potential is finite outside the well, the wavefunction does not vanish but falls exponentially. In other words, the particle can be found in a classically forbidden region that requires greater energy than the energy corresponding to the stationary state. The ground state oscillates least, and it is symmetric with no nodes. As we get to higher and higher states, the nodes increase by one for consecutive states. The energy $E_n$ is smaller because the corresponding wavefunction curves slower as it spreads outside the well and $\psi''(x)$ has smaller magnitude, or it oscillates with smaller frequency. There are only a finite number of bound states, since a state with $E > 0$ is not a bound state.

**Exercise 2.3.1:** Show that the stationary bound state wavefunctions of a system in one dimension are unique (or nondegenerate).

**Solution:** Suppose there are two independent states $\psi_1(x)$ and $\psi_2(x)$ of energy $E$; i.e.,

$$-\frac{\hbar^2}{2m}\frac{d^2\psi_1(x)}{dx^2} + V(x)\psi_1(x) = E\psi_1(x), \quad -\frac{\hbar^2}{2m}\frac{d^2\psi_2(x)}{dx^2} + V(x)\psi_2(x) = E\psi_2(x). \tag{2.3.1}$$

Multiplying the first by $\psi_2(x)$, the second by $\psi_1(x)$ and subtracting,

$$0 = \psi_2(x)\frac{d^2\psi_1(x)}{dx^2} - \psi_1(x)\frac{d^2\psi_1(x)}{dx^2} = \frac{d}{dx}\left(\psi_2(x)\frac{d\psi_1(x)}{dx} - \psi_1(x)\frac{d\psi_1(x)}{dx}\right). \tag{2.3.2}$$

Thus, $\psi_2(x)\frac{d\psi_1(x)}{dx} - \psi_1(x)\frac{d\psi_1(x)}{dx}$ is a constant. Because $\psi_1(x)$ and $\psi_2(x)$ must be nor-

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malizable, they need to vanish at infinity, the constant must be zero. Therefore, $\psi_2(x)\frac{d\psi_1(x)}{dx} - \psi_1(x)\frac{d\psi_1(x)}{dx} = 0$; i.e.,

$$\int \frac{d\psi_1(x)}{\psi_1(x)} = \int \frac{d\psi_2(x)}{\psi_2(x)}. \tag{2.3.3}$$

We then have $\ln(\psi_2(x)) - \ln(\psi_1(x))$ is a constant, or

$$\psi_2(x) = C\psi_1(x), \tag{2.3.4}$$

where $C$ is a constant. Therefore, $\psi_1(x)$ and $\psi_2(x)$ are equivalent and the wavefunction is unique.

**Exercise 2.3.2:** Show that the stationary bound states of a symmetric potential in one dimension must be symmetric or antisymmetric and not a choice.

**Answer:** We know that $\psi(x)$ and $\psi(-x)$ are solutions of the Schrödinger equation with the same energy. Because the wavefunction is unique and nondegenerate in one dimension, the two could only differ by a constant. Therefore, $\psi(-x) = C\psi(x)$. Taking $x \to -x$ again, $\psi(x) = C\psi(-x)$. Thus, $\psi(x) = C^2\psi(x)$ and $C^2 = 1$, which sets $C = \pm 1$. Therefore, either $\psi(-x) = \psi(x)$ and $\psi(x)$ is symmetric or $\psi(-x) = -\psi(x)$ and $\psi(x)$ is antisymmetric.

Now let us consider a particle of mass $m$ in a potential well of finite depth,

$$V(x) = \begin{cases} -V_0, & -a \le x \le a \\ 0, & |x| > a. \end{cases}$$

The potential is shown in the figure below.

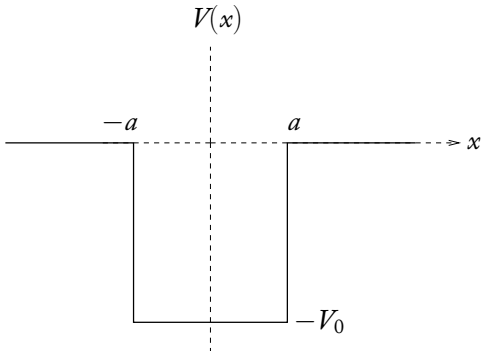



Figure 2.3.1: Finite square well.

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Our interest here is to find the stationary bound states. Because the potential is piecewise and discontinuous, let us call the region on the left where $x < -a$ region *I*, that with in the well $-a \leq x \leq a$ region *II*, and the region where $x > a$ region *III*. The Schrödinger equation in region *I* is

$$\psi_I''(x) - \kappa^2 \psi_I(x) = 0, \qquad \kappa \equiv \sqrt{-\frac{2mE}{\hbar^1}}, \tag{2.3.5}$$

where $E < 0$ for a bound state. The general solution is $\psi_I(x) = Ae^{-\kappa x} + Be^{\kappa x}$. But because the wavefunction has to vanish at $x = -\infty$, in order for it to be normalizable, we set $A = 0$, and we have

$$\psi_I(x) = Be^{\kappa x}. \tag{2.3.6}$$

The Schrödinger equation and the general solution in region *III* are exactly the same as in region *I*. The only difference is that the wavefunction in region *III* must vanish at $x = \infty$, which sets $B = 0$, and we have

$$\psi_{III}(x) = Ae^{-\kappa x}. \tag{2.3.7}$$

The Schrödinger equation in region *II* is

$$\psi_{II}''(x) + k^2 \psi_{II}(x) = 0, \qquad k \equiv \sqrt{\frac{2m(V_0 + E)}{\hbar^1}}, \tag{2.3.8}$$

Because we can choose the solutions to be real and because we know that the bound states will be either even or odd, it is convenient to choose the general solution as a linear combination of sin and cos rather than complex exponentials, and we write

$$\psi_{II}(x) = C \sin kx + D \cos kx. \tag{2.3.9}$$

$C = 0$ and $D \neq 0$ correspond to even solutions, while $D = 0$ and $C \neq 0$. Let us investigate each case separately.

### 2.3.1 Even bound states

Even bound states with $\psi(-x) = \psi(x)$ pick out

$$\psi_{II}(x) = D \cos kx. \tag{2.3.10}$$

Because the potential has only a finite discontinuity at $x = \pm a$, both the wavefunction and its first derivative must be continuous at $x = \pm a$. Continuity of $\psi(x)$ at $x = a$, $\psi_{II}(a) = \psi_{III}(a)$,

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imposes

$$D\cos ka = Ae^{-\kappa a}. \tag{2.3.11}$$

Continuity of $\psi'(x)$ at $x = a$, $\psi'_{II}(a) = \psi'_{III}(a)$, imposes

$$-kD\sin ka = -\kappa Ae^{-\kappa a}. \tag{2.3.12}$$

Imposing continuity of $\psi(x)$ and $\psi'(x)$ at $x = -a$ does not add anything new; it reproduces the same constraints. Dividing the above two, we eliminate the constant coefficients, and we have

$$k\tan ka = \kappa. \tag{2.3.13}$$

Observe that given $a$ and $V_0$, this is a constraint on the allowed energies of the bound states. However, this is a transcendental equation that does not have analytic solutions. We can solve it numerically or graphically as we discuss below. Using the definitions of $k$ and $\kappa$ and rewriting, $\tan ka = \frac{\kappa}{k}$, or $\tan\left(\sqrt{\frac{2m(V_0+E)a^2}{\hbar^2}}\right) = \sqrt{\frac{-E}{V_0+E}} = \sqrt{\frac{-E-V_0+V_0}{V_0+E}} = \sqrt{-1+\frac{V_0}{V_0+E}}$; i.e.,

$$\tan\left(\sqrt{\frac{2m(V_0+E)a^2}{\hbar^1}}\right) = \sqrt{\frac{V_0}{V_0+E}-1}. \tag{2.3.14}$$

Defining new variables

$$z \equiv \sqrt{\frac{2m(V_0+E)a^2}{\hbar^2}}, \qquad z_0 \equiv \sqrt{\frac{2mV_0a^2}{\hbar^2}}, \tag{2.3.15}$$

the constraint becomes

$$\tan z = \sqrt{(\frac{z_0}{z})^2 - 1}. \tag{2.3.16}$$

The solutions can be obtained by plotting each side of (2.3.16) and finding the intersection points as shown in the figure below; i.e. the values of $z$ for which both sides are equal.

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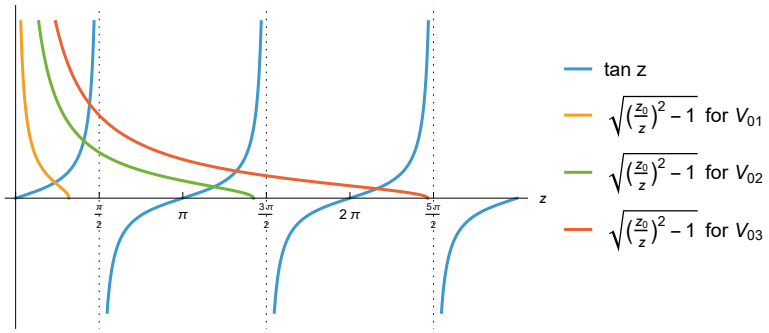



Figure 2.3.2: Graphic solution for the even bound states of the finite square well with depth $V_{01} < V_{02} < V_{03}$.

In the figure above, the potential $V_{01}$ produces only one bound state, since there is only one intersection point, $V_{02}$ produces two bound state, and $V_{03}$ produces three bound state. We also see that any finite well in one dimension supports at least one bound state.

### 2.3.2 Odd bound states

For the odd states,

$$\psi_{II}(x) = C \sin kx. \tag{2.3.17}$$

Continuity of $\psi(x)$ at $x = -a$, $\psi_{II}(-a) = \psi_I(-a)$ imposes

$$C \sin ka = Be^{-\kappa a}. \tag{2.3.18}$$

Continuity of $\psi'(x)$ at $x = -a$, $\psi'_{II}(-a) = \psi'_I(-a)$, imposes

$$kC \cos ka = -\kappa Be^{-\kappa a}. \tag{2.3.19}$$

Dividing the above two,

$$k \cot ka = -\kappa. \tag{2.3.20}$$

Rewriting this in terms of $z$ and $z_0$ as in the even solutions, the constraint becomes

$$\cot z = -\sqrt{(\frac{z_0}{z})^2 - 1}. \tag{2.3.21}$$

This is also a transcendental equation which we solve using graphical method as shown in the figure below

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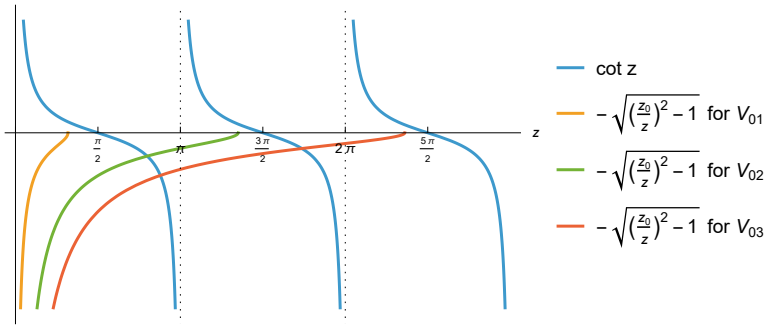



Figure 2.3.3: Graphic solution for the odd bound states of the finite square well with depth $V_{01} < V_{02} < V_{03}$.

In the figure above, the potential $V_{01}$ does not support an odd bound state, since there is no intersection point, $V_{02}$ produce only one bound state, and $V_{03}$ produces two bound state. It means that a finite well potential supports at lease one even state, the ground state, and it may not support any of the excited states.

### 2.3.3 Wide and deep well

**Question 2.3.2:** A wider well of the same depth supports (A). the same number of bound states. (B). larger number of bound states. (C). smaller number of bound states. (D). the same or less number of bound states. (E). the same or more number of bound states.

**Answer:** The answer is (E). the same or more number of bound states. A wider well provides more room for the waves to oscillate providing the possibility for supporting more states. However, the width has to increase by enough amount to support more states. Mathematically, a wider the well gives larger value of $z_0$, and $\sqrt{(\frac{z_0}{z})^2-1}$ or the $-\sqrt{(\frac{z_0}{z})^2-1}$ curves start form higher up and could cross $\tan z$ or $-\cot z$ more number of times. Although factors of $a$ in the ratio $\frac{z_0}{z}$ cancel out, the transcendental equation involves $z$ by itself; i.e., $\tan z = \sqrt{(\frac{z_0}{z})^2-1}$ or $-\cot z = \sqrt{(\frac{z_0}{z})^2-1}$, and we should think of $z_0$ and $z$ as independent parameters in finding the intersection points. More simply, the wider the well, the lower and the closer the energy levels, accommodating equal or more numbers of bound states. What determines the energy is the frequency of oscillation (or how the wavefunction curves), and we can fit excited states with more nodes but of same frequency of oscillation or energy as in a narrower well in a wider well.

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Now suppose the width and depth of the well are such that $z_0 = \sqrt{\frac{2mV_0a^2}{\hbar^2}} >> 1$. In this case, the curve $\sqrt{(\frac{z_0}{z})^2 - 1}$ for the odd case begins high above the $z-$axis and intersects $\tan z$ at its asymptotic values of $\frac{\pi}{2}$, $\frac{3\pi}{2}$, etc, as shown in the figure below.

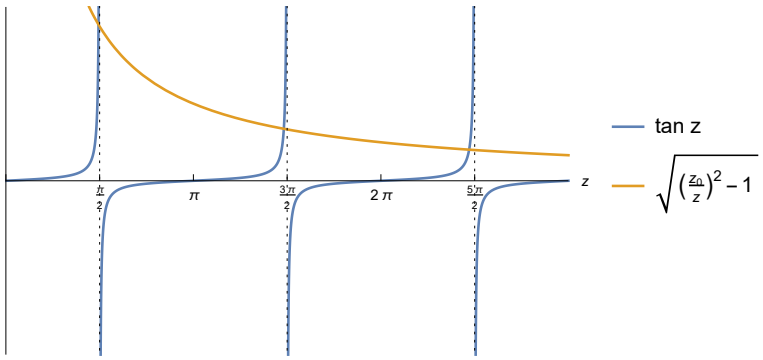



Figure 2.3.4: Graphic solution for the even bound states of a wide and deep finite square well.

Therefore, the intersection points for the odd states are at $z_n \approx \frac{n\pi}{2}$, n=1,3,5,... Similarly for the odd solutions, $z_n \approx \frac{n\pi}{2}$, $n = 2, 4, 6, \ldots$ Combining both the even and odd cases,

$$z_n \approx \frac{n\pi}{2}, \quad n = 1, 2, 3, \ldots \tag{2.3.22}$$

Using the definition of $z$, $\sqrt{\frac{2m(V_0+E)a^2}{\hbar^2}} \approx \frac{n\pi}{2}$, and solving for $E$,

$$E_n = -V_0 + \frac{\hbar^2\pi^2n^2}{8ma^2}, \quad n = 1, 2, 3, \ldots \tag{2.3.23}$$

Thus, the energy levels of the infinite square well of width $2a$ are reproduced for $E_n - (-V_0) = E_n + V_0$, since the ground state energy is shifted down and begins near $E = -V_0$ for the given potential.

### 2.3.4 Shallow and narrow well

We have already seen that any shallow or narrow well in one dimension supports at least one bound state. But we need at least $z_0 = \sqrt{\frac{2mV_0a^2}{\hbar^2}} \geq \frac{\pi}{2}$ to support any odd state, as seen in Figure 2.3.3.

> **Question 2.3.3:** Why is the ground state energy of a bound particle in a potential well not zero, or why is the particle never at rest?

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**Answer:** We can see it from two angles. 1. Using the Schrödinger equation: Because the particle is represented by a wavefunction that is nonzero and oscillates, we have $\psi''(x) \neq 0$ and, following the Schrödinger equation, $E \neq 0$. A solution with $\psi(x) = 0$ represents no particle, and the wavefunction cannot be zero everywhere. 2. Using the uncertainty principle: Because the position of the particle has uncertainty, $\Delta x \neq 0$, and following the uncertainty principle so is the momentum, $\Delta p \neq 0$. Therefore, the particle has momentum and its energy is nonzero. In other words, a quantum particle in a potential well is never at rest, and the ground energy is nonzero.

**Question 2.3.4:** Why are the energy levels of the finite square well smaller than the corresponding energy levels of the infinite square well?

**Answer:** We can see it from the profile of the wavefunctions. The wavefunctions of the finite well vary more gently with $x$ than the corresponding wavefunction of the infinite square well, where it curves more strongly as it vanishes at the boundaries. Therefore, the wavefunctions of the finite square well oscillate slower (or have lower frequency). Or $\psi''(x)$ is smaller than for the corresponding state of the infinite square well. Alternatively, we have seen that the values of $z$ where the points of intersection in the graphical solutions occur at larger values of $z$ which give larger $E$ for larger $V_0$, and in the limit the well is infinitely deep, we reproduce the energies of the infinite square well. If we need more intuition, we know that even the shallowest finite well supports at least one bound state. In the limit the depth is near zero, the energy level supported is near zero too, which is smaller than the ground state energy of the infinite well. The pattern continues for the higher energy levels too, and as the depth increases the lower lying levels and the corresponding wavefunction become closer to that of the infinite square well.

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